

Probability
Chapter 4 - Module 1

Introduction to Markov Chains

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Def. (Stochastic Process) A stochastic process is a collection of random variables $X(t)$ for $t \in T$.

The set T is called the index set and $t \in T$ an index. We use the symbol T because in many cases the index set is related to time. For example we might have $T = \{1, 2, 3, \dots\}$ or $T = \{0, 1, 2, 3, \dots\}$ or $T = [0, \infty]$.

For each t , $X(t)$ is called the state for index t and we often denote the set of all states as S , so that for each $t \in T$, $X(t) \in S$.

Technically $X(t)$ is a random variable and an observed value of $X(t)$ would be a state in S .

Example 1. Let $X(i), i = 1, 2, \dots$ be an infinite sequence of iid Bernoulli random variables with parameter p , the probability of success.

More interesting cases are when the stochastic process represents a sequence of dependent events.

Example 2. Let $Y_n = \sum_{i=1}^n X_i$. Then Y_n is a binomial random variable with mean np . Clearly these random variables are dependent, since $Y_{n+1} = Y_n + X_{n+1}$ and information about Y_n provides information about Y_{n+1} . In particular

$$E[Y_{n+1} \mid Y_n = k] = k + E[X_{n+1}] = k + p,$$

whereas using linearity of the expectation operator,

$$E[Y_{n+1}] = (n+1)p.$$

Example 3. (Left Skip Free Random Walk)

As before let $X_1, X_2, \dots$ be a sequence of iid random variables with

$$P(X_i = j) = P_j$$

for $i = 1, 2, \dots$ and where $j \in \{-1, 0, 1, \dots\}$.

Then as in Example 2., the sequence $S_n = \sum_{i=1}^n X_i$ is a sequence of dependent random variables.

Example 4. (Random Walk) Let $X_0 = 0$ and for each $i = 0, 1, \dots$ let

$$X_{i+1} = \begin{cases} X_i + 1 & \text{with probability } p \\ X_i - 1 & \text{with probability } 1 - p. \end{cases}$$

In this example $T = \{0, 1, \dots\}$ and the state space S is the set of integers.

To further simplify the presentation we will assume in future cases (unless otherwise noted) that the state space is $S = \{0, 1, \dots\}$ so that $X_n = i$ for some $i \in \{0, 1, \dots\}$ for $n \in \{0, 1, \dots\}$

Def. (Markov Chain) Let $X_0, X_1, \dots$ be a finite or countable infinite sequence of random variables. We say that $\{X_i\}_{i=0}^{\infty}$ is a Markov Chain if

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i).$$

We then define

$$P_{ij} = P(X_{n+1} = j \mid X_n = i).$$

These are called the transition probabilities for the Markov Chain. When these are stored in a matrix P with elements P_{ij} , the finite or possible infinite matrix P is called the matrix of transition probabilities.

Note especially that P_{ij} does not depend on n .

Theorem. Let P be the matrix of transition probabilities. Then

$$\sum_{j=0}^{\infty} P_{ij} = 1.$$

Proof.

$$\begin{aligned} \sum_{j=0}^{\infty} P_{ij} &= \sum_{j=0}^{\infty} P(X_{n+1} = j \mid X_n = i) \\ &= P(X_{n+1} \in S \mid X_n = i) \\ &= P(X_{n+1} \in S) \\ &= 1. \end{aligned}$$

Example 4. (revisited) In the random walk example,

$$X_{i+1} = \begin{cases} X_i + 1 & \text{with probability } p \\ X_i - 1 & \text{with probability } 1 - p \end{cases}$$

so that

$$P(X_{n+1} = j + 1 \mid X_n = j) = p$$

$$P(X_{n+1} = j - 1 \mid X_n = j) = 1 - p$$

and

$$P(X_{n+1} = j \mid X_n = i) = 0$$

otherwise.

Example 3. (revisited) For the Left Skip Free Random Walk

$$P(X_i = j) = P_j, \quad j = -1, 0, \dots$$

with

$$S_0 = 0$$

and

$$S_n = \sum_{k=1}^n X_k.$$

So

$$P(S_{n+1} = j \mid S_n = i, S_{n-1} = i_{n-1}, \dots, S_0 = i_0) = P(S_{n+1} = j \mid S_n = i)$$

so that

$$P_{i,i+j} = P(S_{n+1} = i + j \mid S_n = i) = P_j$$

for $j = -1, 0, \dots$ and zero otherwise.

Chapman-Kolmogorov Theorem. Let P^n denote the n -step transition probability matrix. Then $P_{ij}^{n+m} = \sum_{k=1}^{\infty} P_{ik}^n P_{kj}^m$.

Proof.

$$\begin{aligned} P_{ij}^{n+m} &= P(X_{n+m} = j \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P(X_{n+m} = j, X_n = k \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P(X_{n+m} = j \mid X_n = k, X_0 = i) P(X_n = k \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P(X_{n+m} = j \mid X_n = k) P(X_n = k \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P(X_m = j \mid X_0 = k) P(X_n = k \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P_{kj}^m P_{ik}^n \\ &= \sum_{k=0}^{\infty} P_{ik}^n P_{kj}^m. \end{aligned}$$

We originally defined P_{ij}^n as the probability of moving from state i to state j in exactly n steps. We can represent this as a matrix.

What the Chapman-Kolmogorov Theorem shows is that P^n really is the transition matrix P raised to the n^{th} power!

Example 4.10 An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces.

If initially both balls are red, find the probability that the fifth ball selected is red.

We solve this in the following way.

$P(\text{fifth selection is red})$

$$\begin{aligned} &= \sum_{i=0}^2 P(\text{fifth selection is red} \mid X_4 = i) P(X_4 = i \mid X_0 = 2) \\ &= 0 P_{2,0}^4 + .5 P_{2,1}^4 + 1 P_{2,2}^4. \end{aligned}$$

Let's compute this using Mathematica.

Example 5. Let's return to a classic problem, which is that we flip a sequence of fair coins and let N denote the number of flips until there is a run of 4 consecutive heads. What is the probability that $N = 10$? What is the probability that $N \leq 10$. What is the probability that $N = 100$? Can you estimate or maybe even compute these probabilities?

We turn this into a Markov chain problem in the following way.

Let $X_0 = 0$ and for each $n \geq 1$ let X_n denote that length of the current run of heads.

So for example if we have the following outcomes: HTTHHTHHH, the

$$X_1 = 1, X_2 = X_3 = 0, X_4 = 1, X_5 = 2, X_6 = 0.$$

For $i = 0, 1, 2, 3$, $X_n = 0$ with probability $1/2$ and increases by 1 with probability $1/2$.

And we define $P(X_{n+1} = 4 \mid X_n = 4) = 1$ and $P(X_{n+1} = j \mid X_n = 4) = 0$ for $j = 0, 1, 2, 3$.

Let's call up Mathematica and look at the matrix of transition probabilities.

The long term behaviour of a Markov chain is in some cases independent of the initial state. To get to that result and to understand the structure of Markov chains we need several new ideas.

Def. A state j is said to be **accessible** from a state i if $P_{ij}^n > 0$ for some $n \geq 0$.

Which is to say that if you are in state i , then it is possible for the chain to eventually enter state j . Note that if j is accessible from state i , it is not necessarily true that state i is accessible from state j .

Def. Two states i and j are said to **communicate** with each other if state j is accessible from state i and state i is and accessible from state j .

The relation i communicates with state j is an equivalence relation in that it satisfies the following three conditions.

1. For each i , state i communicates with state i . (Satisfied by $n = 0$.)
2. If state i communicates with state j , then state j communicates with state i . (Follows immediately from the definition.)
3. If state i communicates with state j and state j communicates with state k , then state i communicates with state k .

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Item 3. follows from the fact that if i communicates with state j , then for some n , $P_{ij}^n > 0$. And if state j communicates with state k , then for some m , $P_{jk}^m > 0$. It then follows that

$$P_{ik}^{n+m} = \sum_{r=0}^{\infty} P_{ir}^n P_{rk}^m \geq P_{ij}^n P_{jk}^m > 0$$

from which we conclude that state k is accessible from state i . A similar argument shows that state i is accessible from state k .

So state i communicates with state k .

So “communicates with” is an equivalence relation on the set of all states. This defines a set of equivalence classes and within each class, every state communicates with every other state. And states in distinct equivalence classes do not communicate with each other.

Def. A Markov chain is said to be irreducible if there is only one class.

Question: Consider the following transition matrix. Does this come from an irreducible Markov chain?

$$P = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & 2/3 \end{pmatrix}$$

And what is the answer for the following transition matrix?

$$P = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 1/2 & 1/2 & 0 \\ 1/3 & 2/3 & 0 \end{pmatrix}$$

Def. For any state i , let f_i denote the probability that, starting in state i , that the process will ever re-enter state i .

State i is said to be recurrent if $f_i = 1$ and transient if $f_i < 1$.

If state i is recurrent, then the probability of eventually revisiting that state is 1. But once you visit a state, since this is a Markov chain, the probability that you will revisit the state again is 1. So one expects that the probability that you will revisit that state infinitely often is 1.

On the other hand, if state i is transient, then there is a positive probability, $1 - f_i$, that you will never revisit that state again. Furthermore, the probability that one will only visit that state exactly n times is $f_i^n(1 - f_i)$, for $n \geq 1$.

But we recognize this as the geometric random variable with mean $\frac{1}{1 - f_i}$ and this is the expected number of times that one will revisit this state.

We conclude that if a state i is recurrent, then given that we are in state i , the expected number of times that the process is in that state is infinite.

$$\text{Let } I_n = \begin{cases} 1, & \text{if } X_n = i \\ 0. & \text{if } X_n \neq i. \end{cases}$$

Then

$$\begin{aligned} E \left[\sum_{n=1}^{\infty} I_n \mid X_0 = i \right] &= \sum_{n=1}^{\infty} E[I_n \mid X_0 = i] \\ &= \sum_{n=1}^{\infty} P(X_n = i \mid X_0 = i) \\ &= \sum_{n=1}^{\infty} P_{ii}^n. \end{aligned}$$

Theorem. State i is

$$\text{recurrent if } \sum_{n=1}^{\infty} P_{ii}^n = \infty$$

$$\text{transient if } \sum_{n=1}^{\infty} P_{ii}^n < \infty.$$

Corollary 4.2. If state i is recurrent, and state i communicates with state j , then state j is recurrent.

It follows from this that if state i is transient and communicates with state j , then state j is transient, for if not, then state j would be recurrent, which would imply that state i is recurrent, which it is not.

Theorem. In any irreducible Markov chain with a finite number of states, every state is recurrent.

Proof. An irreducible Markov chain is one in which every state communicates with every other state. So since there is just one class, every state must either be recurrent or every state must be transient.

But since there are only a finite number of states, some state must be visited infinitely many times and it is not possible that every state is transient. So some state must be recurrent. Therefore, every state is recurrent.